

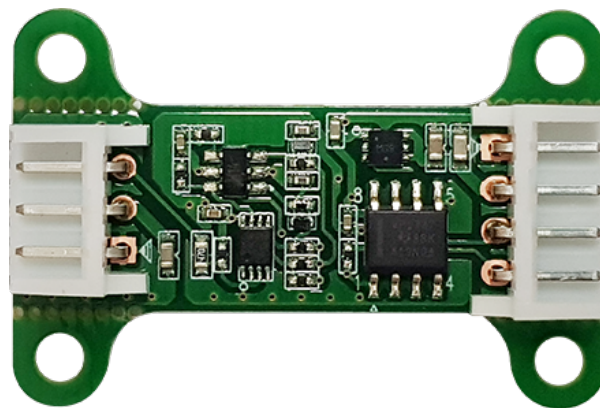


DXL Communication Bridge

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1.

Introduction

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The **DYNAMIXEL Communication Bridge** works as a communication converter between TTL (3 Pin JST Type connector) and RS485 (4 Pin JST Type connector) of DYNAMIXEL.

[ROBOTIS OpenSourceTeam](#)

NOTE

TTL and RS485 ports in the [U2D2 interface](#) are separately connected to the master PC, however, the data line between TTL and RS485 ports are not connected to each other.



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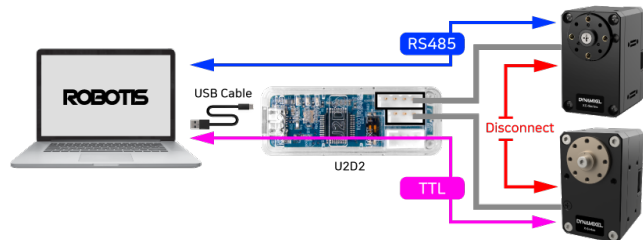
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Thus, Instructions Packets such as **Broadcast Ping, Sync Read and Bulk Read** over TTL to RS485 and vice versa are not feasible through the U2D2 interface only.

- For more details about Instruction Packet, see [DYNAMIXEL Protocol 1.0](#) or [Protocol 2.0](#) depending on what DYNAMIXEL is in use.



Unlike the U2D2, the DYNAMIXEL Communication Bridge features a signal conversion between TTL to RS485, which allows Broadcast Ping, Sync Read, and Bulk Read instructions from mixed types of communication.



2. Specifications

Item	Specifications
Operating Voltage	3.5 V ~ 30.0 V
Current Rating	3.0 A
Baud Rate	Max 4 Mbps



DANGER

(Ignoring these warnings may cause serious injury or death)

- Never place items containing water, flammables/ open flames, or solvents near the product.
- Never place fingers, arms, toes, and other body parts near product during operation.
- Cease operation and remove power from the product if the product begins to emit strange odors, noises, or smoke.



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CAUTION

(Ignoring these warnings may cause mild injury or damage to the product)

- Always comply with the product's official operating environment specifications including input voltage, current, and operating temperature.
- Do not insert blades or other sharp objects during product operation.



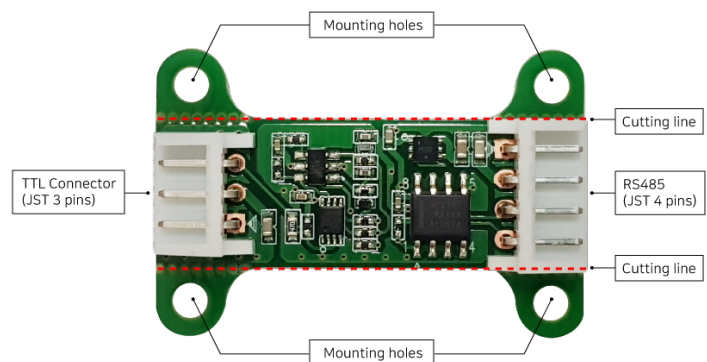
ATTENTION

(Ignoring these warnings may cause minor injury or damage to the product)

- Do not disassemble or modify the product.
- Do not drop the product or apply strong impacts.
- Do not connect or disconnect DYNAMIXEL cables while power is being supplied.

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3. Layout



Layout - Front

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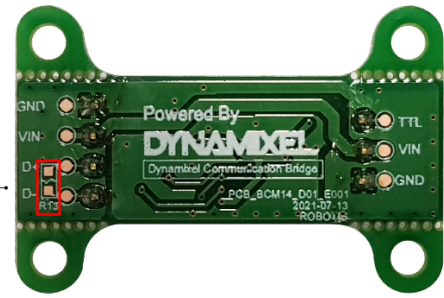
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RS485 termination resistance



Layout - Back

TOP

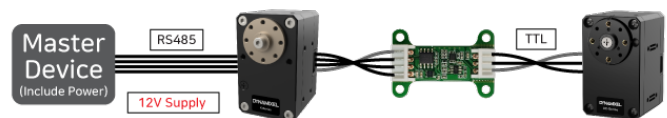
NOTE: **VDD** is the same terminology as the **Vin** printed back of the DYNAMIXEL Communication Bridge.

3. 1. Description

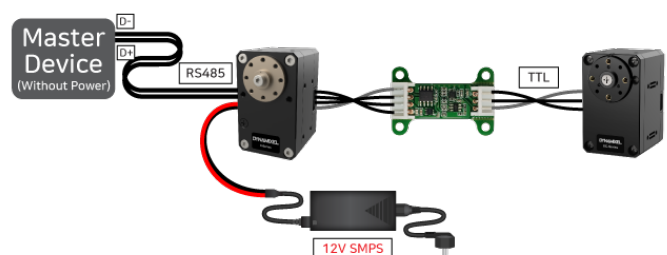
- 1. TTL (JST 3 Pins) / RS485 Connecters (JST 4 Pins) :** 3 and 4 Pin JST connectors for TTL / RS458 communication.
- 2. Mounting holes :** Used to mount on other devices. Mounting holes can be removed if not used by snapping along the V-cut line.
- 3. RS485 Termination resistance: Reserved space to install RS485 termination resistor.** 1608 size(1.6 x 0.8 mm) chip resistor can be used.

4. Connection Examples

- If the operating voltages of all DYNAMIXELs are the same, see the following images.



Powering the DYNAMIXEL from Master Device.





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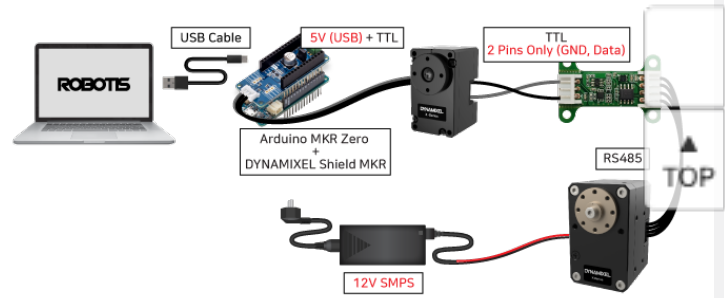
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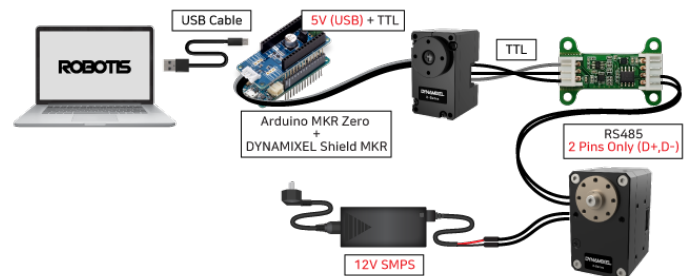
5. Reference

Powering the DYNAMIXEL from external power source.

- If the operating voltages of DYNAMIXELs are different, see the following images.



Powering the DYNAMIXEL Communication Bridge from the RS485 connector side.



Powering the DYNAMIXEL Communication Bridge from the TTL connector side.

WARNING: **VDD** pins (See [Layout](#)) of DYNAMIXEL Communication Bridge between a RS485 and TTL connector are connected each other. In case the TTL and RS485 connectors use different power, be sure to connect **only one VDD** to the DYNAMIXEL Communication Bridge for the product safety.

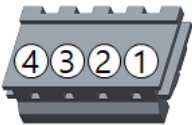
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5. 1. Connector Information

Item	TTL	RS-485
Pinout	1 GND 2 VDD 3 DATA	1 GND 2 VDD 3 DATA+ 4 DATA-
Diagram		

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Item	TTL	RS-485
Housing	 JST EHR-03	 JST EHR-04
PCB Header	 JST B3B-EH-A	 JST B4B-EH-A
Crimp Terminal	JST SEH-001T-P0.6	JST SEH-001T-P0.6
Wire Gauge for DYNAMIXEL	21 AWG	21 AWG

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